

PATRICK BUROG CARPIO

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PROFESSIONAL SUMMARY

Computer Engineering student specializing in Artificial Intelligence with hands-on experience developing AI-powered applications, machine learning systems, and mobile applications. Skilled in Python, Flutter, FastAPI, SQL, and machine learning frameworks, with a focus on AI engineering and intelligent systems.

EDUCATION

2023 - Present

Batangas State University Alangilan Campus | Alangilan, Batangas City, Batangas

Bachelor of Science in Computer Engineering, Specialization in Artificial Intelligence

Related Courses: Programming, Logic and Design, Object Oriented Programming, Data Structures and Algorithms, Machine Learning, Deep Learning, Software Design

Expected Graduation: July 2027

RELATED EXPERIENCE AND PROJECTS

AI-Powered Overdue Invoice Collection System

Full-Stack Developer | Personal Project

- Developed an AI-assisted invoice collection system that analyzes customer payment behavior and predicts payment likelihood using Python, XGBoost/Random Forest, and scikit-learn, with a dynamic three-stage escalation workflow for automated and personalized follow-ups.
- Built a full-stack AI architecture using Flutter, FastAPI, PostgreSQL, Hugging Face Transformers, and Ollama, incorporating NLP-based response classification, payment-plan recommendations, human approval, and a feedback loop for continuous model improvement.

Notiq: Intelligent Mobile Study Companion

Full-Stack Developer | Personal Project

- Developed an AI-powered Flutter study companion that converts handwritten and digital study materials into structured reviewers, active-recall flashcards, and automatically scored quizzes using Google ML Kit, Syncfusion Flutter PDF, docx-to-text, and Llama 3.3 70B via the Groq API.
- Built a local-first study management system using SQLite, Provider, and local notifications, featuring note and library management, exam scheduling, progress and accuracy tracking, interactive dashboards, dark mode, and automated deadline reminders.

SmartFit: A Wearable IoT System for Real-Time Exercise Form Evaluation, Posture Monitoring, and Exertion Detection Using IMU and Heart Rate Sensors

Ongoing Thesis for 3rd and 4th year

- Developing a wearable IoT fitness system using dual ESP32 nodes, IMU and heart-rate sensors, ESP-NOW, and Bluetooth Low Energy (BLE) to provide real-time exercise form, posture, exertion, and heart-rate monitoring through a Flutter Android application.
- Implementing AI-assisted fitness coaching and local data processing using the Groq API, Flutter, Riverpod, SQLite, and BLE, with features including body-scan analysis, goal assessment, personalized exercise and meal recommendations, form alerts, workout summaries, and progress tracking.

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TECHNICAL SKILLS

Programming: *Python, Java, C++, C#, Dart, JavaScript, TypeScript, SQL, HTML, CSS*

AI / Machine Learning: *Scikit-learn, XGBoost, Random Forest, YOLO, Hugging Face Transformers, Ollama*

Frameworks: *Flutter, FastAPI, Flask, React, Next.js*

Databases: *PostgreSQL, SQLite, Firebase, Supabase*

Tools & Platforms: *Git, GitHub, VS Code, Google Colab, Figma, Vercel, Heroku*

CERTIFICATES

CCNA: Introduction to Networks

December 2025

Cisco Networking Academy

HTML Fundamentals

August 2026

CodeCred